

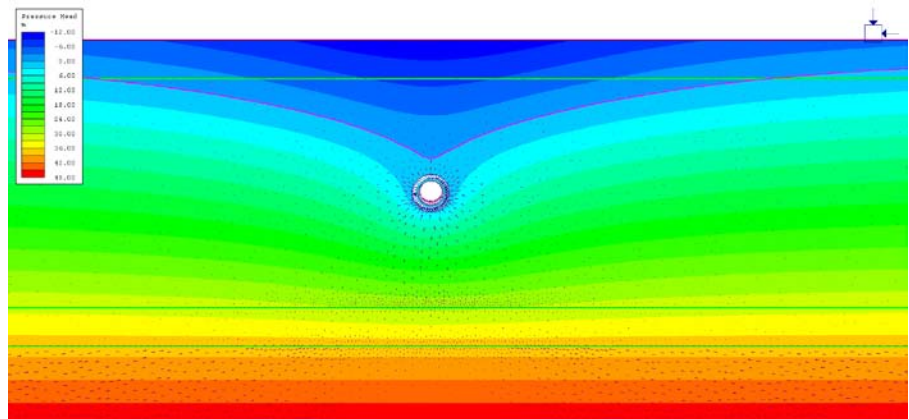
Drawdown due to tunnel excavation

In this tutorial, *Phase2* is used to simulate tunnel excavation in a saturated soil. The tunnel is assumed to have a permeable liner so there is a drawdown in the water table as water drains into the tunnel. The model is based on a study presented by Shin et al. (2002).

The complete model can be found in the **Tutorial 27 Tunnel Drawdown.fez** file. All tutorial files installed with *Phase2* 8.0 can be accessed by selecting File > Recent Folders > Tutorials Folder from the *Phase2* main menu.

Topics covered

- Staged tunnel excavation
- Groundwater drawdown
- Permeable tunnel liner
- User defined permeability function
- 3D tunnel simulation



Model

Start the *Phase2* Model program.

Project Settings



Open the **Project Settings** dialog from the **Analysis** menu and make sure the General tab is selected. Define the units as being “Metric, stress as kPa”. Click on the tab for Stages. Change the number of stages to 3.

Project Settings

General
Stages
Stress Analysis
Groundwater
Statistics
Strength Reduction
Project Summary

Stages

#	Name
1	Stage 1
2	Stage 2
3	Stage 3

Number of Stages: 3

Edit Stages:

* Inserted stages are shown in red

OK Cancel

Click on the tab for Groundwater. Set the Method to Finite Element Analysis. Leave all other options as default.

Project Settings

General
Stages
Stress Analysis
Groundwater
Statistics
Strength Reduction
Project Summary

Groundwater

Method: Finite Element Analysis

Pore Fluid Unit Weight: 9.81 kN/m3

Grid Interpolation: Modified Chugh

Compute:
☐ Groundwater Only
☒ Groundwater AND Stress

Maximum Number of Iterations: 500

Tolerance: 1e-006

OK Cancel

Click OK to close the Project Settings dialog.

Geometry

The problem consists of four soil layers and an excavation. Therefore an external boundary, three material boundaries, and an excavation boundary are required.



Start by creating a rectangular external boundary. Select the **Add External** option in the **Boundaries** menu and enter the following coordinates:

-100 , 0

100 , 0

100 , -50

-100 , -50

c (to close the boundary)



Now we need to delineate the different material layers within the model. Go to the **Boundaries** menu and select **Add Material**. Enter the following points:

-100 , -5

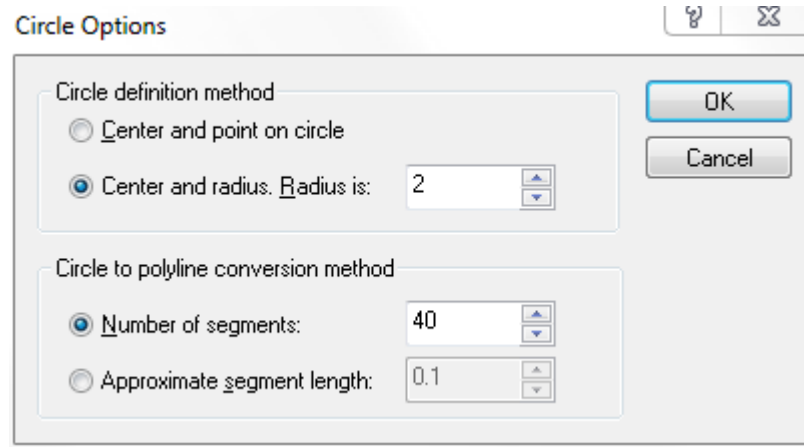
100 , -5

Hit Enter to finish entering points. All of the material layers are horizontal, so we will simply copy this boundary. Right click on the green material boundary and select Copy Boundary. Click on the point at (-100,-5). Enter -100 , -35 at the prompt and hit Enter. Repeat, entering -100 , -40 for the second point. The model should look like this:

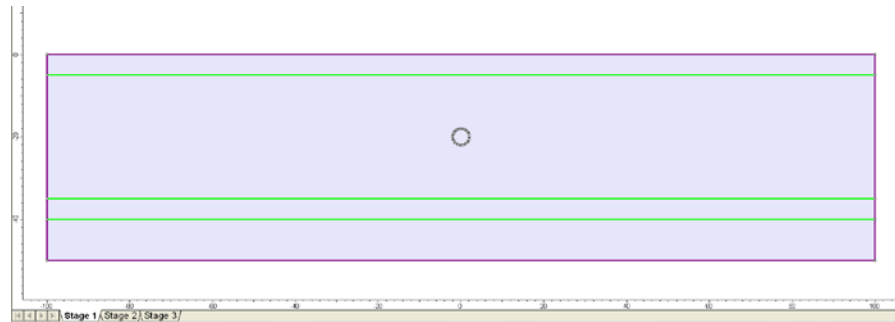




Now we will create the tunnel. From the **Boundaries** menu, select **Add Excavation**. Type i followed by Enter to create a circular boundary. In the resulting dialog, choose the option 'Centre and Radius'. Set the radius to 2 m. Leave the Number of Segments as 40.



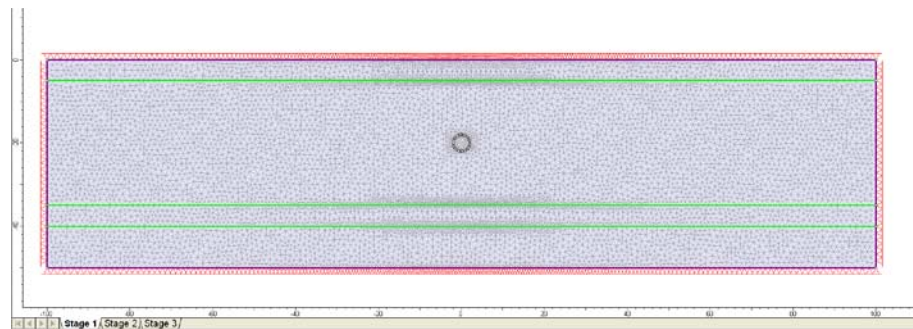
Click OK to close the dialog. Enter the coordinates 0 , -20 for the centre of the circle. Hit Enter. The model should now look like this:



Mesh



Generate the finite element mesh by selecting **Discretize and Mesh** from the **Mesh** menu. The mesh will look like this:

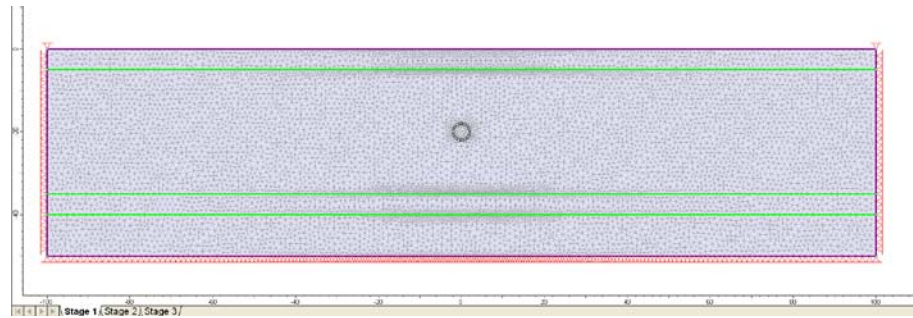


Boundary conditions



By default, the entire external boundary is fixed. Since the top of this model represents the actual ground surface, we need to free the top surface. Go to the **Displacements** menu and select **Free**. Click on the ground surface and hit Enter. You will now see that the fixed boundary conditions have disappeared from the top boundary.

We now need to re-establish the fixed boundary condition on the top corners. Right click on the top left corner and select Restrain X,Y. Repeat for the top right corner. Your model should now look like this:



Field Stress



Because the top of the model represents the true ground surface, we want to use a gravity field stress. Go to the **Loading** menu and select **Field Stress**. For Field Stress Type select Gravity and click the check box for “Use actual ground surface”. Leave all other values as default.

Field Stress Properties

Field Stress Type: **Gravity**

☒ Use actual ground surface ☐ Use effective stress ratio ☐ Use variable stress ratio

Ground Surface Elevation (m):

Unit Weight of Overburden (kN/m3):

Total Stress Ratio (horiz/vert in plane):

Total Stress Ratio (horiz/vert out-of-plane):

Locked-in horizontal stress (in plane) (kPa, Comp. +):

Locked-in horizontal stress (out-of-plane) (kPa, Comp. +):

OK **Cancel** **Statistics...** **Advanced >>**

Click OK to close the dialog.

Materials



We now need to define the material properties and assign the correct materials to the correct parts of the model. Go to the **Properties** menu and select **Define Materials**. Change the name of Material 1 to Thames gravels. Enter the material parameters as shown.

The screenshot shows the 'Define Material Properties' dialog box for 'Thames gravels'. The left pane lists materials from 'Thames gravels' to 'Material 28'. The right pane shows the following settings:

- Name:** Thames gravels
- Material Color:** (Color selection box)
- Initial Element Loading:** Field Stress & Body Force
- Unit Weight: (kN/m3):** 20
- Elastic Properties:**
 - Elastic Type:** Isotropic
 - Poisson's Ratio:** 0.3
 - Young's Modulus (kPa):** 100000
 - Young's Modulus (resid) (kPa):** 20000
 - E1 (kPa):** 20000
 - E2 (kPa):** 20000
 - Ez (kPa):** 20000
 - v12:** 0.2
 - v1z:** 0.2
 - v2z:** 0.2
- Strength Parameters:**
 - Failure Criterion:** Mohr Coulomb
 - Material Type:** Plastic
 - Tensile Strength (peak) (kPa):** 0
 - Dilation Angle (deg):** 17.5
 - Fric. Angle (peak) (deg):** 35
 - Fric. Angle (resid) (deg):** 35
 - Cohesion (peak) (kPa):** 0
 - Cohesion (resid) (kPa):** 0
 - Tensile Strength (resid) (kPa):** 0
- Stage Properties:** (unchecked)
- Datum Dependent:** (unchecked)
- Unsaturation Shear Strength:**
 - Phi b:** 0
 - Air Entry (kPa):** 0

Buttons at the bottom: Copy To..., Statistics..., Show only properties used in model, OK, Cancel.

Click on the tab for Material 2, change the name to London clay and enter the following properties:

Define Material Properties

☐ Thames gravels

☐ London clay

☐ Lambeth Group clay

☐ Lambeth Group Sand

☐ Material 5

☐ Material 6

☐ Material 7

☐ Material 8

☐ Material 9

☐ Material 10

☐ Material 11

☐ Material 12

☐ Material 13

☐ Material 14

☐ Material 15

☐ Material 16

☐ Material 17

☐ Material 18

☐ Material 19

☐ Material 20

☐ Material 21

☐ Material 22

☐ Material 23

☐ Material 24

☐ Material 25

☐ Material 26

☐ Material 27

☐ Material 28

London clay

Name: London clay

Material Color:

Initial Element Loading: Field Stress & Body Force

Unit Weight: (kN/m3): 20

Elastic Properties

Elastic Type: Isotropic

Poisson's Ratio: 0.3

Young's Modulus (kPa): 7000

Young's Modulus (resid) (kPa): 20000

E1 (kPa): 20000

E2 (kPa): 20000

Ez (kPa): 20000

v12: 0.2

v1z: 0.2

v2z: 0.2

Strength Parameters

Failure Criterion: Mohr Coulomb

Material Type: Plastic

Tensile Strength (peak) (kPa): 0

Dilation Angle (deg): 0

Fric. Angle (peak) (deg): 23

Fric. Angle (resid) (deg): 23

Cohesion (peak) (kPa): 5

Cohesion (resid) (kPa): 5

Tensile Strength (resid) (kPa): 0

☐ Stage Properties

☐ Datum Dependent

Unsat. Shear Strength

Phi b: 0

Air Entry (kPa): 0

Define Factors...

Define Properties...

Copy To...

Statistics...

☐ Show only properties used in model

OK

Cancel

Enter the following properties for materials 3 and 4.

Define Material Properties

Lambeth Group clay

Name: Lambeth Group clay Material Color:

Initial Element Loading: Field Stress & Body Force Unit Weight: (kN/m3): 20

Elastic Properties

Elastic Type: Isotropic Poisson's Ratio: 0.3

Young's Modulus (kPa): 20000 ☐ Young's Modulus (resid) (kPa): 20000

E1 (kPa): 20000 E2 (kPa): 20000 E3 (kPa): 20000

ν_{12} : 0.2 ν_{13} : 0.2 ν_{23} : 0.2

Strength Parameters

Failure Criterion: Mohr Coulomb Material Type: Plastic

Tensile Strength (peak) (kPa): 0 Dilation Angle (deg): 0

Fric. Angle (peak) (deg): 27 Fric. Angle (resid) (deg): 27

Cohesion (peak) (kPa): 25 Cohesion (resid) (kPa): 25

Tensile Strength (resid) (kPa): 25

☐ Stage Properties ☐ Datum Dependent

Define Factors... Define Properties...

Unsaturated Shear Strength

Phi b: 0 Air Entry (kPa): 0

Copy To... Statistics... ☐ Show only properties used in model OK Cancel

Define Material Properties

Lambeth Group Sand

Name: Lambeth Group Sand Material Color:

Initial Element Loading: Field Stress & Body Force Unit Weight: (kN/m3): 20

Elastic Properties

Elastic Type: Isotropic Poisson's Ratio: 0.3

Young's Modulus (kPa): 20000 ☐ Young's Modulus (resid) (kPa): 20000

E1 (kPa): 20000 E2 (kPa): 20000 E3 (kPa): 20000

ν_{12} : 0.2 ν_{13} : 0.2 ν_{23} : 0.2

Strength Parameters

Failure Criterion: Mohr Coulomb Material Type: Plastic

Tensile Strength (peak) (kPa): 0 Dilation Angle (deg): 17.5

Fric. Angle (peak) (deg): 34 Fric. Angle (resid) (deg): 34

Cohesion (peak) (kPa): 0 Cohesion (resid) (kPa): 0

Tensile Strength (resid) (kPa): 0

☐ Stage Properties ☐ Datum Dependent

Define Factors... Define Properties...

Unsaturated Shear Strength

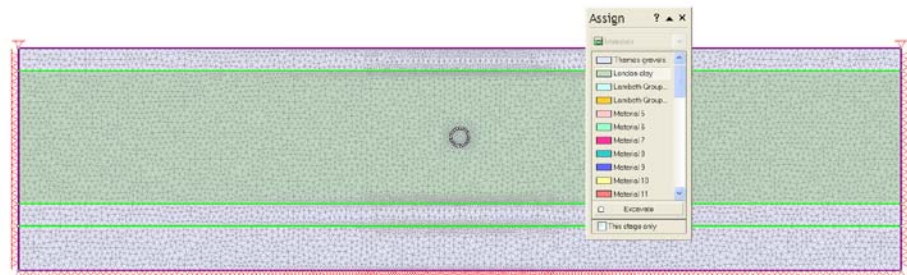
Phi b: 0 Air Entry (kPa): 0

Copy To... Statistics... ☐ Show only properties used in model OK Cancel

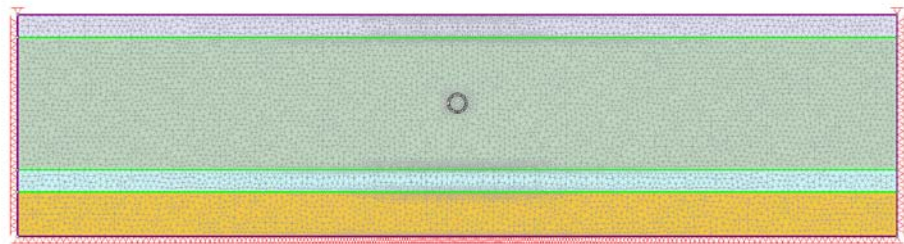
Click OK to close the dialog.



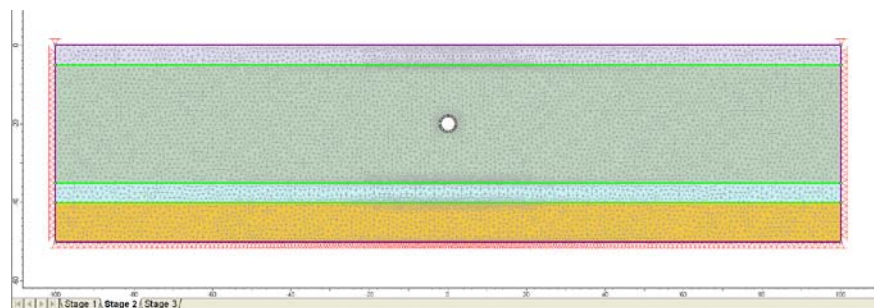
To assign the materials to the model, select **Properties → Assign Properties**. Be sure you are looking at the first stage. By default, everything is set to Thames gravel material. Select London clay from the assign dialog and click in the second layer. Be sure to click inside the tunnel as well.



Now choose Lambeth Group clay and click in the third layer, and Lambeth Group sand and click in the fourth layer. The model should look like this:



Click on the Stage 2 tab near the bottom left of the window. Choose Excavate from the Assign menu and click inside the tunnel. Close the Assign dialog. The model will now look like this:



Excavation and support

In general, a tunnel must be excavated before structural support is added. Therefore, some deformation will occur before the support is installed. The amount of deformation depends on the distance of the tunnel face from the location of support installation. The tunnel face provides some support, so if the liner is installed close to the tunnel face, then not much deformation will have occurred. In essence, this is a three-dimensional problem that we are trying to simulate in two dimensions.

For this problem, let's assume that we are installing the liner 2 m behind the face. To simulate the supporting effect of the nearby tunnel face, we will apply a load to the perimeter of the tunnel equal to some proportion of the in-situ stress. The amount of load required to correctly simulate the 3D effect of the tunnel face can be determined by following procedures outlined in Tutorial 24.

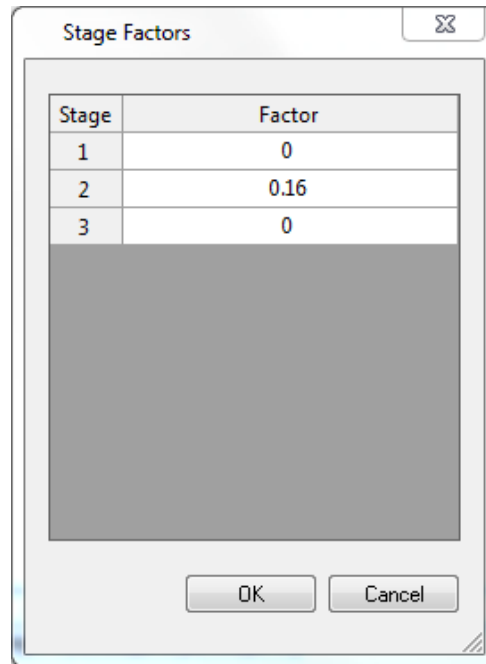
In the third stage, we will install the liner and remove the load. Removing the load will simulate the advance of the tunnel face away from our two-dimensional slice.



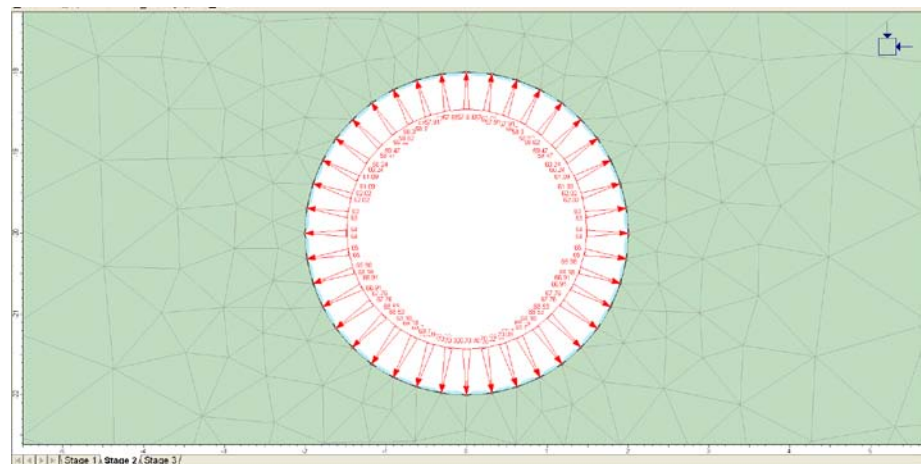
Ensure you are looking at Stage 2. Go to **Loading** → **Distributed Loads** → **Add Uniform Load**. Under Orientation, choose the option for Field Stress Vector. This will automatically work out the traction required to perfectly balance the in-situ stress. Now click the checkbox for Stage Load.

The screenshot shows the 'Add Distributed Load' dialog box. It has a title bar with a help icon and a close icon. The 'Orientation' section contains several radio buttons: 'Normal to boundary', 'Vertical (+ve down)', 'Horizontal (+ve right)', 'Angle from horizontal', 'Angle to boundary', and 'Field stress vector' (which is selected). Below these is a small coordinate system icon and a text box with '0'. To the right, the 'Magnitude (kN/m2)' is set to '0'. There is a checked checkbox for 'Stage Load' and a 'Stage Factors...' button. At the bottom are 'OK' and 'Cancel' buttons.

Click on the Stage Factors button. Through the type of analysis described in Tutorial 24, we can determine that a load of 0.16 times the in-situ stress will simulate the effective support of the tunnel face 2 m away. Therefore for Stage 2, set the Stage Factor to 0.16. For Stage 3, we assume that the tunnel face is far away so set the Stage Factor to 0.

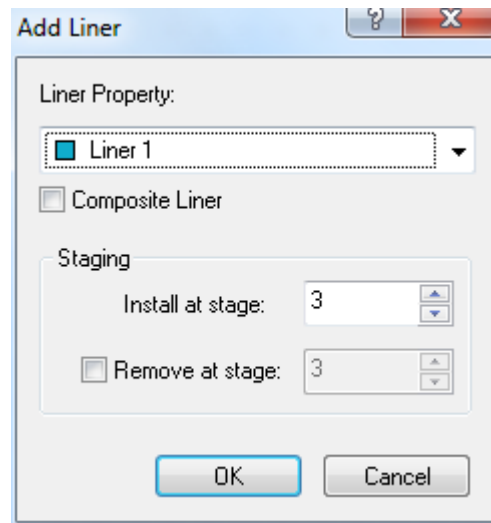


Click OK to close the dialog. Click OK to close the 'Add Distributed Load' dialog. You will now be prompted to select a boundary on which to apply the load. Click somewhere above and to the left of the tunnel. Hold down the left mouse button and draw a window around the tunnel. Release the left mouse button and hit Enter to select the tunnel boundary. Zoom in using the middle mouse wheel, or click the Zoom Excavation button. The model for Stage 2 should now look like this:

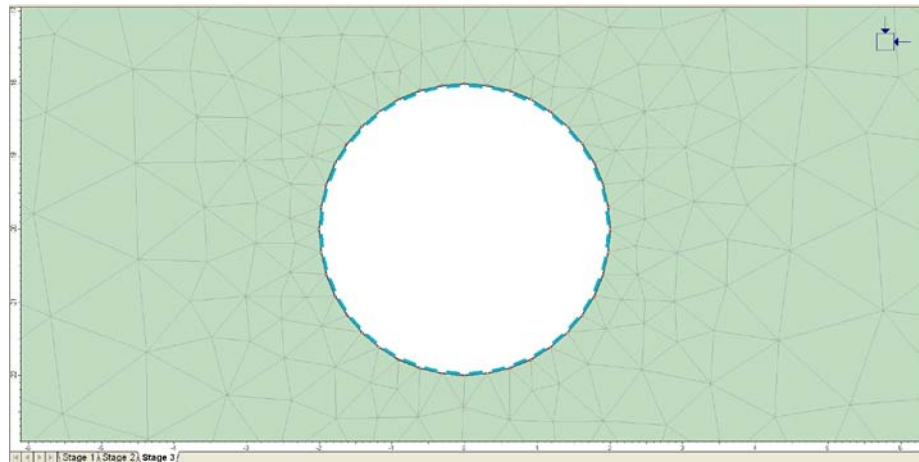


You can see how the applied traction is not constant. It is calculated to balance the in-situ stress, which increases with depth.

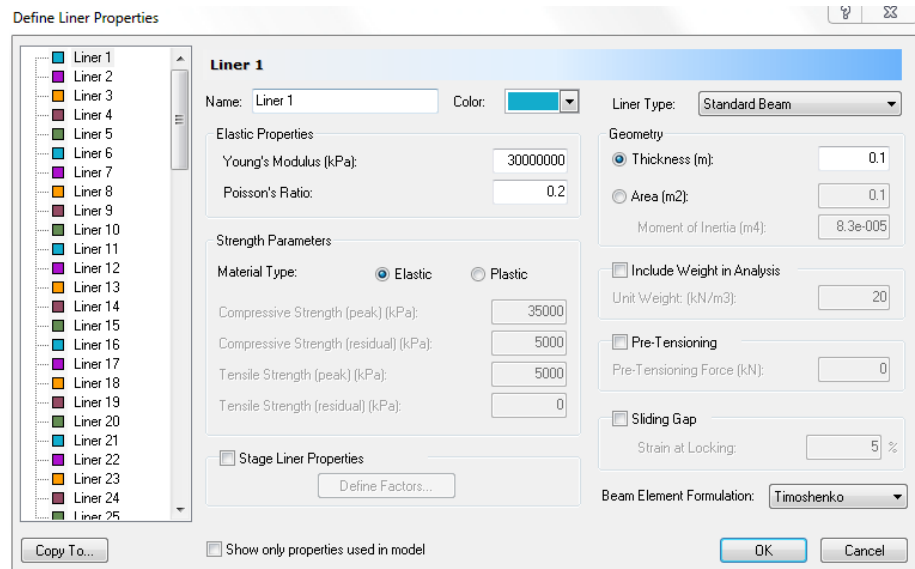
Click on the tab for Stage 3. Go to **Support** → **Add Liner**. Ensure 'Liner Property' is Liner 1 and 'Install at stage' is 3.



Click OK and draw a window around the tunnel as you did when applying loads above. Hit Enter and your model should look like this:



To see the properties of the liner, right click on it and select Liner Properties.



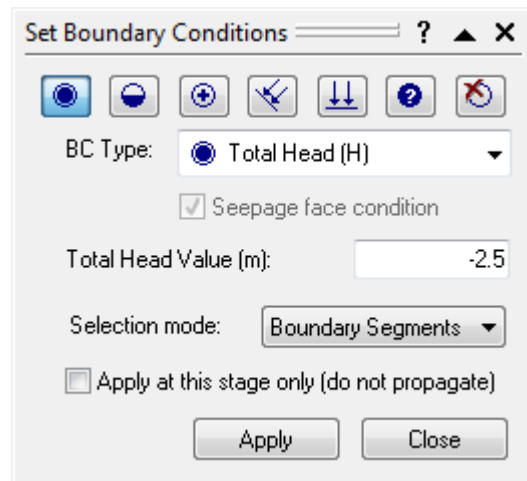
The default values are suitable so click OK to close the dialog.

Groundwater boundary conditions

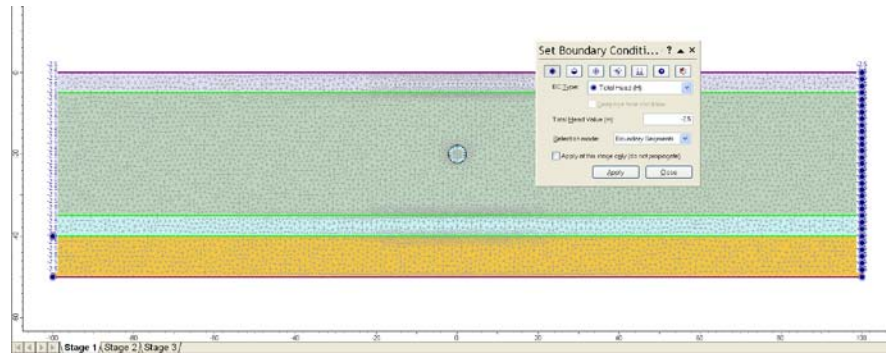
The water table for this site is 2.5 m below the ground surface. We will use groundwater boundary conditions to set this up.



Go back to Stage 1. Zoom out to see the entire model using the mouse wheel or the Zoom All button. From the **Groundwater** menu, select **Show Boundary Conditions**. Choose **Set Boundary Conditions** from the **Groundwater** menu. For the BC Type, choose Total Head. Set the Total Head Value to -2.5 m.

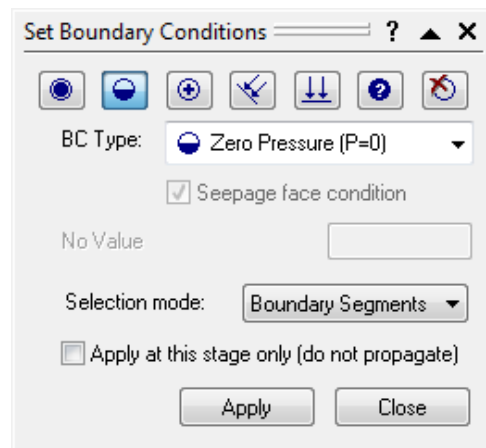


Select all sections of the left and right vertical boundaries. Click Apply. The model will appear as shown:

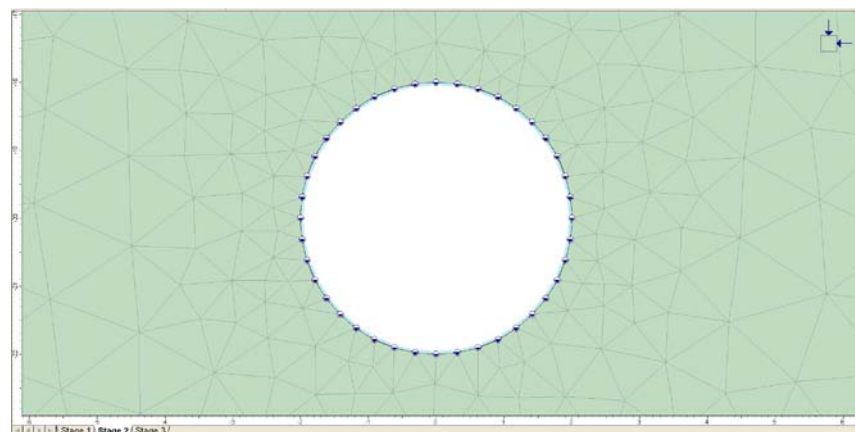


We will now simulate the effect of a permeable liner by setting the pressure at the surface of the tunnel to 0.

Click on the tab for Stage 2. Zoom in on the tunnel. From the Set Boundary Conditions dialog, select BCType = Zero Pressure.



Draw a window around the tunnel as described above. Click the Apply button and then the Close button. The model should look like this:



Click on the tab for Stage 3 to ensure that the zero pressure boundary condition also exists in this stage.

Hydraulic material properties

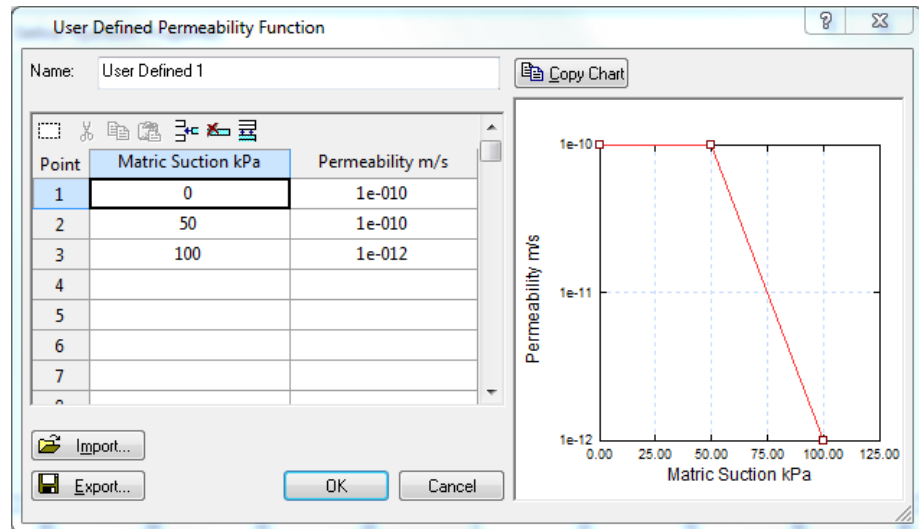


From the **Properties** menu, choose **Define Hydraulic**. Ensure you are looking at the properties for Thames gravel. Here you can select the model that dictates the permeability transition from saturated to unsaturated soil. Leave the model as the default (Simple). You can also change the permeability here. Set the value to $1\text{e-}10$ m/s.

The screenshot shows the 'Define Hydraulic Properties' dialog box for 'Thames gravels'. On the left is a list of materials: Thames gravels, London clay, Lambeth Group clay, Lambeth Group San, Material 5, Material 6, Material 7, Material 8, Material 9, Material 10, Material 11, Material 12, Material 13, and Material 14. The 'Thames gravels' tab is selected. The 'Model' is set to 'Simple'. The 'Hydraulic Parameters' section shows 'Ks' set to $1\text{e-}10$ m/s, 'K2 / K1' set to 1, and 'K1 Angle' set to 0. The 'Simple Parameters' section shows 'Soil Type' set to 'General'. There are buttons for 'New...', 'Plot Function...', 'Copy To...', 'Define Factors...', 'OK', and 'Cancel'. A checkbox for 'Show only properties used in model' is also present.

ASIDE: Obviously, the permeability of gravel is much higher than we have specified. However, in this problem, we want to observe the drawdown of the water table due to the tunnel excavation. Drawdown will only occur if the recharge rate is low, i.e. water does not quickly enter the system to replace the water lost into the tunnel. If we set a permeability for the top layer higher than the underlying layer, then rapid recharge will occur and we will not see drawdown of the water table.

Click on the tab for London clay. The base permeability for the London clay is $1\text{e-}10$ m/s. The permeability decreases by two orders of magnitude between 50kPa and 100 kPa of suction. We can simulate this behaviour with a user-defined permeability model. For Model, click the New button. Set the Name to Clay model and fill in the chart as shown:

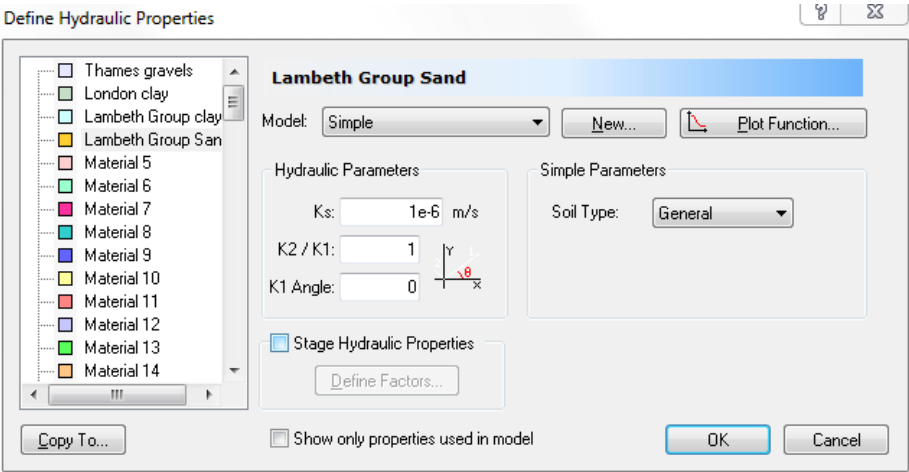


Click OK to return to the Define Hydraulic Properties dialog.

Click on the tab for Lambeth Group clay. Leave the model as Simple and set the permeability to 1e-10 m/s.

The 'Define Hydraulic Properties' dialog box is shown with the 'Lambeth Group clay' tab selected. The 'Model' is set to 'Simple'. The 'Hydraulic Parameters' section shows Ks: 1e-10 m/s, K2 / K1: 1, and K1 Angle: 0. The 'Simple Parameters' section shows Soil Type: General. The 'Stage Hydraulic Properties' section is unchecked. The 'Show only properties used in model' checkbox is also unchecked. The 'Copy To...' button is visible at the bottom left.

Click on the tab for Lambeth Group sand. Set the permeability to 1e-6 m/s as shown.



Click OK to close the dialog.



The model definition is now complete. Save the model using the **Save As** option in the **File** menu.

Compute



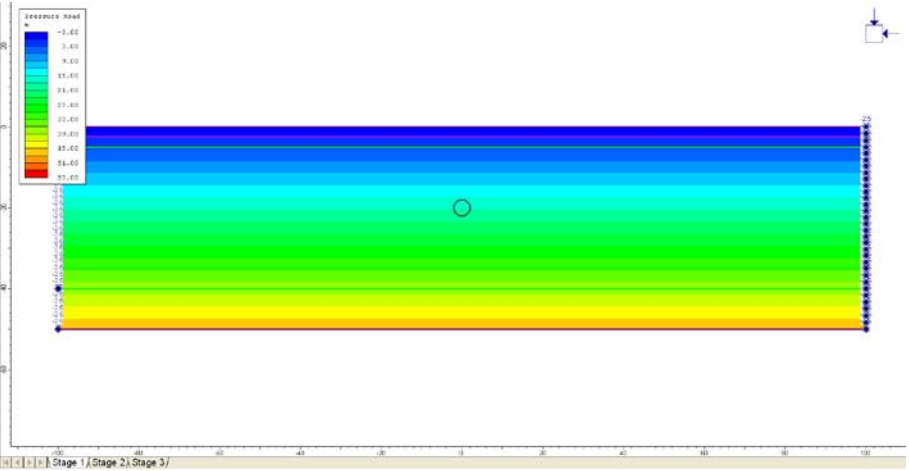
Run the model using the **Compute** option in the **Analysis** menu. The model should take a couple of minutes to compute.



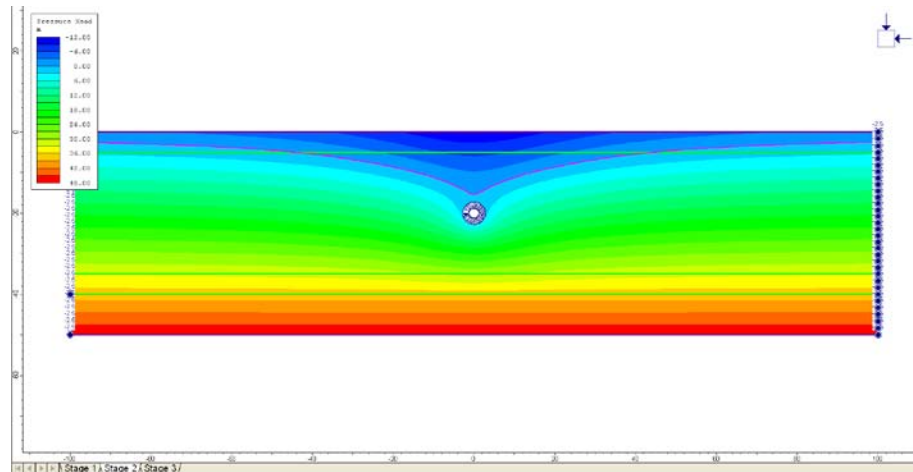
Once the model has finished computing (Compute dialog closes), select the **Interpret** option in the **Analysis** menu to view the results.

Interpret

The Interpret program starts and reads the results of the analysis. You are now looking at the Pressure Head in Stage 1.



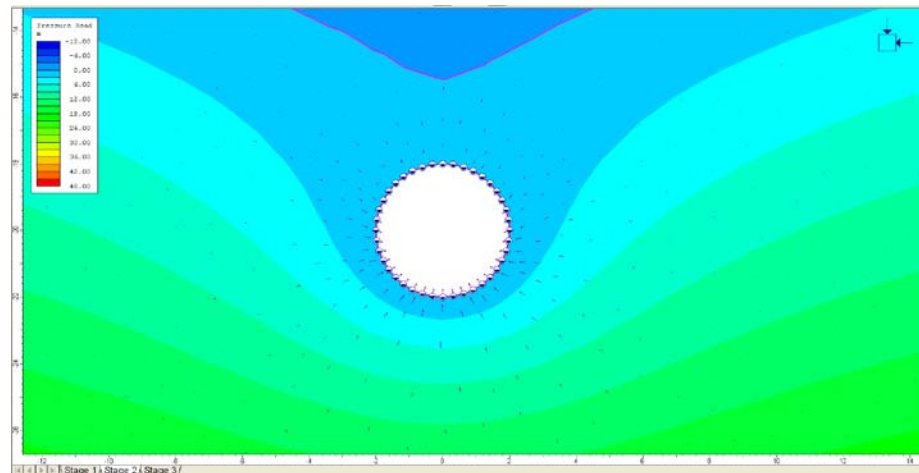
As expected, the water table (pink line) is at 2.5 m below the surface and the pressure increases monotonically with depth. Click on the tab for Stage 2.



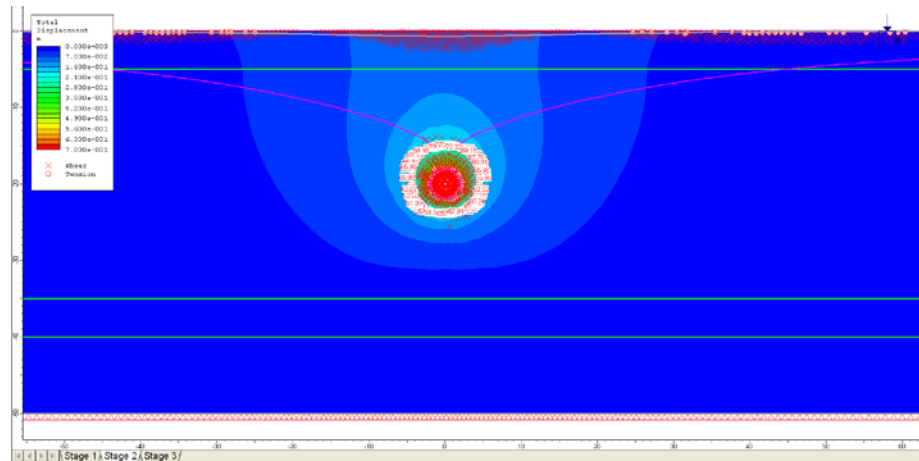
Here you can see the obvious drawdown of the water table due to the drained boundary around the tunnel.



Show the flow vectors by clicking on the Show Flow Vectors button in the toolbar. Zoom in on the tunnel and you can see how the fluid is flowing into the tunnel.



Turn off the Flow Vectors. Change the plot to show Total Displacement contours. Zoom out until you can see the ground surface. Click the button to Display Deformed Boundaries. Click the button to Display Yielded Elements. The plot should look like this:



You can see some shear failure around the tunnel and at the ground surface. You can also see how there is some subsidence at the surface. To determine the exact value, go to **Query** → **Add Material Query**. Enter 0,0 for the query point. Hit Enter. In the resulting dialog, choose At Each Vertex and Show Queried Values.

Specify Query Locations

Generate Values

☐ At locations, evenly spaced along the ENTIRE poly-line

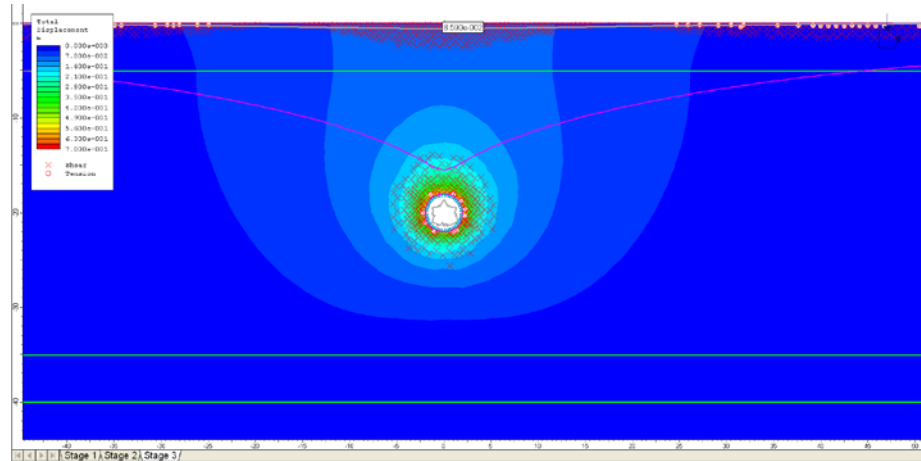
☒ At each vertex

☐ At each vertex, and at points evenly spaced along EACH SEGMENT

☒ Show queried values ☐ Show ID Number

Click OK. You will see that the displacement directly above the tunnel is about 8.8 cm.

Click on the tab for Stage 3. You will see that there is little change in the displacement or failure pattern (there is actually a small amount of rebound since removing the load is equivalent to removing material inside the tunnel).



The liner has successfully accommodated the load without any more failure. To evaluate the performance of the liner, go to **Analysis** → **Show Values** → **Show Values**. Check the box for Liners and choose Bending Moment from the pull-down menu.

Show Values

Data

☐ Joints Normal Stress

☐ Springs Horizontal Force

☐ Bolts Axial Force

☒ Liners Bending Moment

Drawing Style

☐ Positive and negative on same side

☐ Draw on opposite side

☐ Fill

☒ Hatch

Hatch Style:

Positive Colour:

Negative Colour:

Size of largest value (all stages)

☒ 10 [mm] on screen and paper

☐ 1 [m] in model coordinates

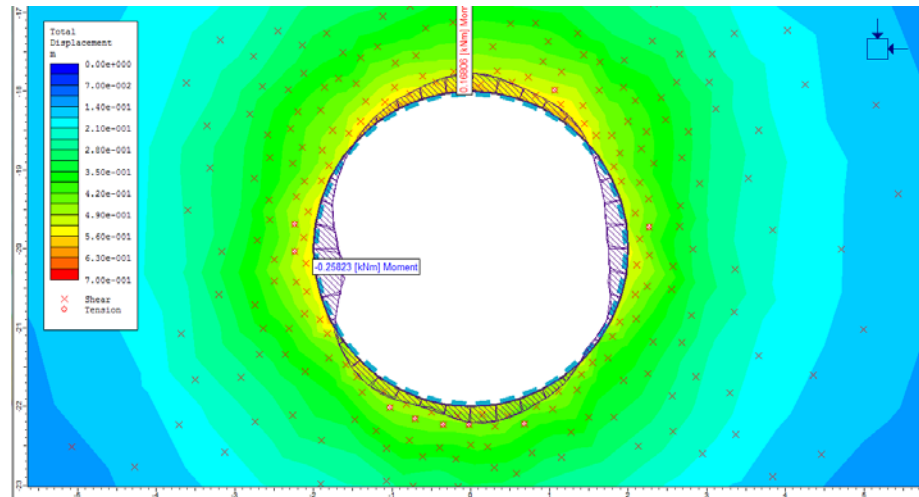
Display

☒ Bars ☒ Lines ☐ All Values

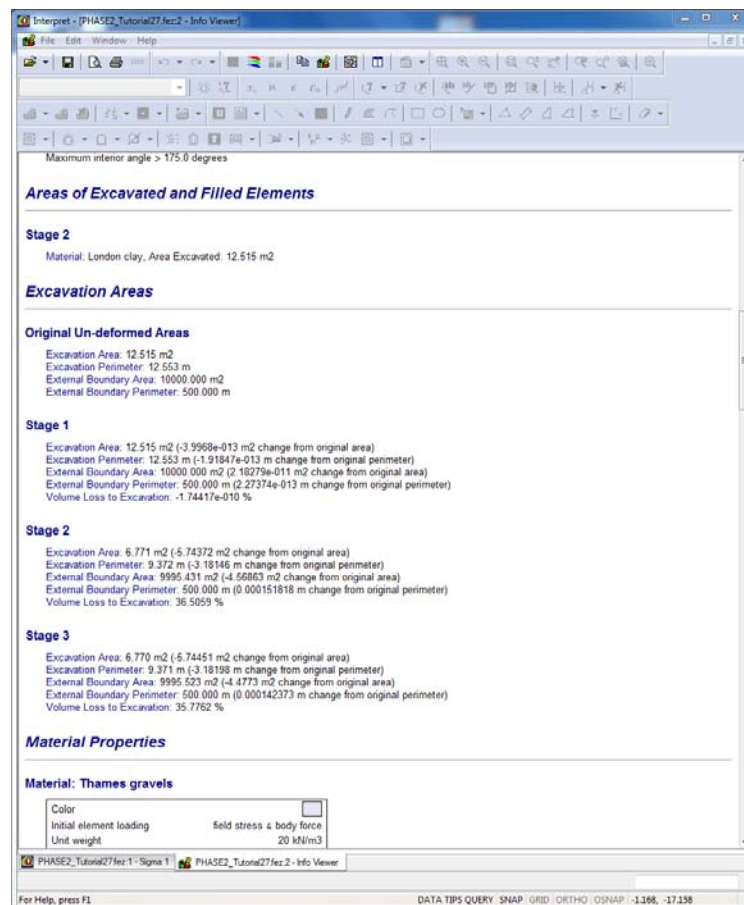
☒ Minimum and Maximum Values

Apply OK Cancel

Click OK. Zoom in on the tunnel to see the moments as shown.



Finally, you can check the volume loss due to the tunnel excavation. The volume loss is the volume change due to surface subsidence divided by the volume of the excavation. Go to **Analysis** → **Info Viewer**. Scroll down to the heading for Stage 3. You can see the Volume Loss to Excavation is about 35 %.



This concludes the tutorial; you may now exit the *Phase2* Interpret and *Phase2* Model programs.

References

Shin, J.H., Addenbrooke, T.I. and Potts, D.M., 2002. A numerical study of the effect of groundwater movement on long-term tunnel behaviour. *Géotechnique*, **52** (6), 391-403.